

Formal Logic (SM234614)

First Meeting: What is Formal Logic?

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What do you know about “Formal Logic”?

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Formal Logic

The abstract study of propositions, statements, or assertively used sentences and of deductive arguments.

<https://www.britannica.com>

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The abstract study of **propositions**, **statements**, or assertively **used sentences** and of **deductive** arguments.

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Formal Logic

Formal logic is logic that deals with the form or logical structure of statements and propositions and the logical implications and relations that exist or come about because of those logical forms. In particular, formal logic is concerned with the forms that yield or guarantee valid inferences from a premise or premises to a conclusion.

<https://www.newworldencyclopedia.org>

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Compare the following statements:

“I am hungry” vs “Eventually, I will be hungry”

Compare the following statements:

“Right now I am hungry” vs “Eventually, I will be hungry”

Compare the following statements:

“Right now I am hungry”	vs	“Eventually, I will be hungry”
(propositional logic)		(temporal logic)

Compare the following statements:

“Right now I am hungry” vs “Eventually, I will be hungry”

(propositional logic)

(temporal logic) temporal logic:

propositional logic + temporal operator

Compare the following statements:

“Right now I am hungry” vs “Eventually, I will be hungry”

(propositional logic)

(temporal logic) temporal logic:

propositional logic + temporal operator

- ▶ Next
- ▶ Until
- ▶ Eventually
- ▶ Always
- ▶ etc..

Formal modeling: to model a “phenomena” in a “formal” way

Formal specification: to express specifications/requirement in a “formal” way

Formal verification: to check whether a model satisfies the underlying specification in a “formal” way

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WORDLE



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V	I	R	A	L
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KATLA #17



K	E	R	J	A
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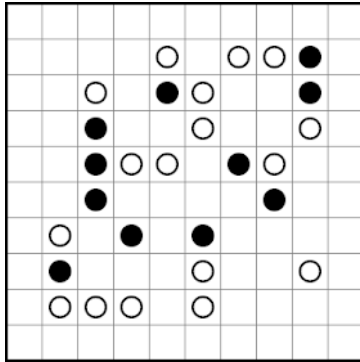
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**FORMAL VERIFICATION AND MODELING OF
STOCKS PRICE USING DISCRETE TIME MARKOV
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AGFA SYALSA FADLINA

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**FORMAL VERIFICATION AND MODELLING OF
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DISCRETE-TIME MARKOV CHAIN AND
PROBABILISTIC COMPUTATION TREE LOGIC**

MOEHAMAD NOOR AT THARIKH

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MARCELLIUS FERNANDO ROBOT

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